

Prompt Library Platform

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Roadmap:

NXUI-84 - Getting issue details... STATUS

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Prompt Library Architecture

Overview

The Prompt Library provides a production-ready platform for managing, discovering, and suggesting high-quality prompts across Adobe Experience Platform applications. Built for enterprise-scale operations, it enables both administrators and end-users to access curated prompt collections that accelerate AI Assistant adoption and ensure consistent, high-quality user interactions. The system includes an administrative interface for prompt management, intelligent search and discovery capabilities, automated similarity detection to prevent duplication, and seamless integration with AI Assistant user interfaces across AEP applications.

Value Proposition

The Prompt Library transforms the AI Assistant user experience by providing:

- **Accelerated Onboarding:** New users discover AI capabilities through curated prompt suggestions
- **Quality Assurance:** Administrative oversight ensures prompts are accurate, relevant, and aligned with product capabilities
- **Reduced Friction:** Users spend less time formulating queries and more time gaining insights
- **Consistency:** Standardized prompts promote best practices and optimal AI Assistant usage patterns
- **Discoverability:** Intelligent search and categorization help users find the right prompt for their task
- **Personalization:** Automatic filtering by role (B2B/B2C) and application ensures relevant suggestions

User Interface

Overview

The Prompt Library provides two distinct user experiences:

1. An administrative interface for prompt management and curation.
2. An end-user interface embedded within AI Assistant experiences across Adobe Experience Platform applications.

Use Case: End-User Prompt Discovery

Business users access the Prompt Library directly within their AI Assistant interface. When opening the AI Assistant chat, users are greeted with suggested prompts tailored to their role, application context, and common tasks.

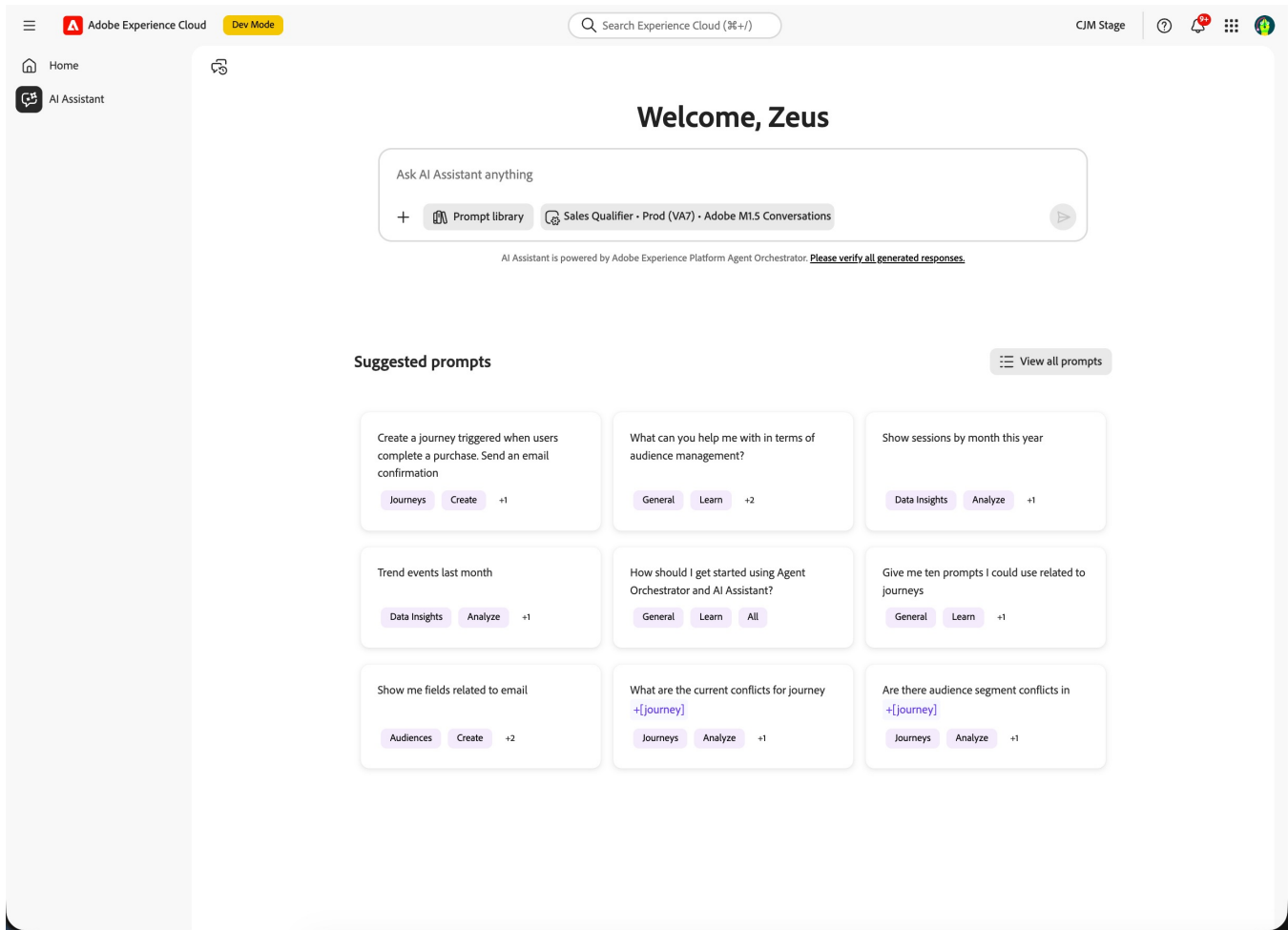


Figure 1: AI Assistant interface displaying personalized prompt suggestions organized by goal and domain with quick access to relevant queries.

Key Features:

- **Contextual Suggestions:** Prompts filtered by user's application (AEP, AJO, CJA)
- **Role-Based Filtering:** Automatic detection of B2B vs B2C customer type
- **Goal-Oriented Organization:** Prompts categorized by intent (Learning, Analyze, Optimize, Troubleshooting, Create)

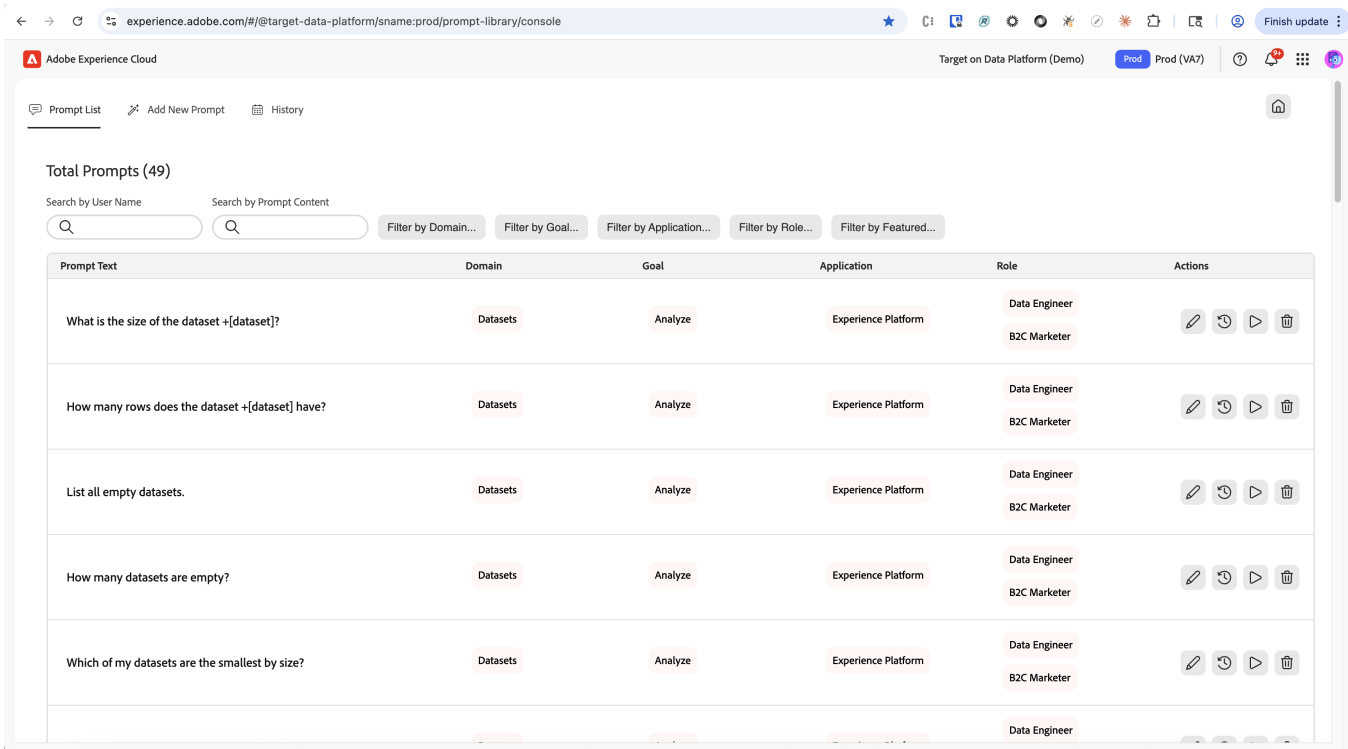
- **Featured Prompts:** Highlighted prompts for common or high-value queries
- **Quick Actions:** One-click prompt insertion into the chat interface

Use Case: Administrative Prompt Management (Admin Tool)

Administrators use the Prompt Library Admin UI to curate, manage, and maintain the prompt catalog. The interface provides comprehensive CRUD operations, version history tracking, and intelligent similarity detection to prevent duplicate prompts.

This SPA was built as an enterprise-grade admin tool with:

- Dedicated repository and independent deployment
- LaunchDarkly feature flag (`prompt-library-admin-ui`) that currently gates the console
- RBAC security



https://adobe-my.sharepoint.com/:v/p/sbiswari/lQCjoilqOna9Q5wjKZnZ-g_sAUaHQdYI-irxLxTFsiwQ2bY?e=Sbwlad

Admin Capabilities:

- **Prompt Catalog Management:** Create, update, and delete prompts with rich metadata
- **Similarity Detection:** Automatic detection of semantically similar prompts before creation
- **Version History:** Complete audit trail of all prompt modifications
- **Featured Promotion:** Mark high-value prompts for prominent display
- **Role & Application Targeting:** Configure prompt visibility by role and application
- **Search & Filter:** Advanced search across prompt text, metadata, and user modifications

Use Case: Similarity Detection for Quality Control

When administrators create or update prompts, the system automatically checks for semantically similar existing prompts. This prevents duplication and helps maintain a clean, high-quality prompt catalog.

Similarity Detection Flow:

1. Administrator drafts a new prompt text
2. System generates semantic embedding and queries existing prompts
3. Similar prompts are surfaced with similarity scores
4. Administrator reviews suggestions and decides to:
 - Proceed with new prompt if sufficiently unique
 - Update existing similar prompt instead
 - Merge concepts into a refined prompt

Definitions

- **Prompt** – A curated query or question template designed to elicit specific AI Assistant capabilities. *Example: "What can you help me with in terms of audience management?"*
- **Goal** – The user's intent or objective when using a prompt. *Examples: Learning, Analyze, Optimize, Troubleshooting, Create.*
- **Domain** – The functional area or product capability addressed by a prompt. *Examples: General, Audiences, Journeys, Data Insights, Experimentation.*
- **Role** – The user's job function or persona. *Examples: B2C Marketer, B2B Marketer, Data Engineer.*
- **Application** – The Adobe Experience Platform product where the prompt is relevant. *Examples: Experience Platform, Adobe Journey Optimizer, Customer Journey Analytics.*
- **Featured Prompt** – A high-value prompt marked for prominent display in user interfaces.
- **Similarity Score** – A numerical measure (0-1) indicating semantic similarity between two prompts, used for duplicate detection.
- **Prompt Version** – A historical snapshot of a prompt at a specific point in time, enabling audit trails and rollback capabilities.

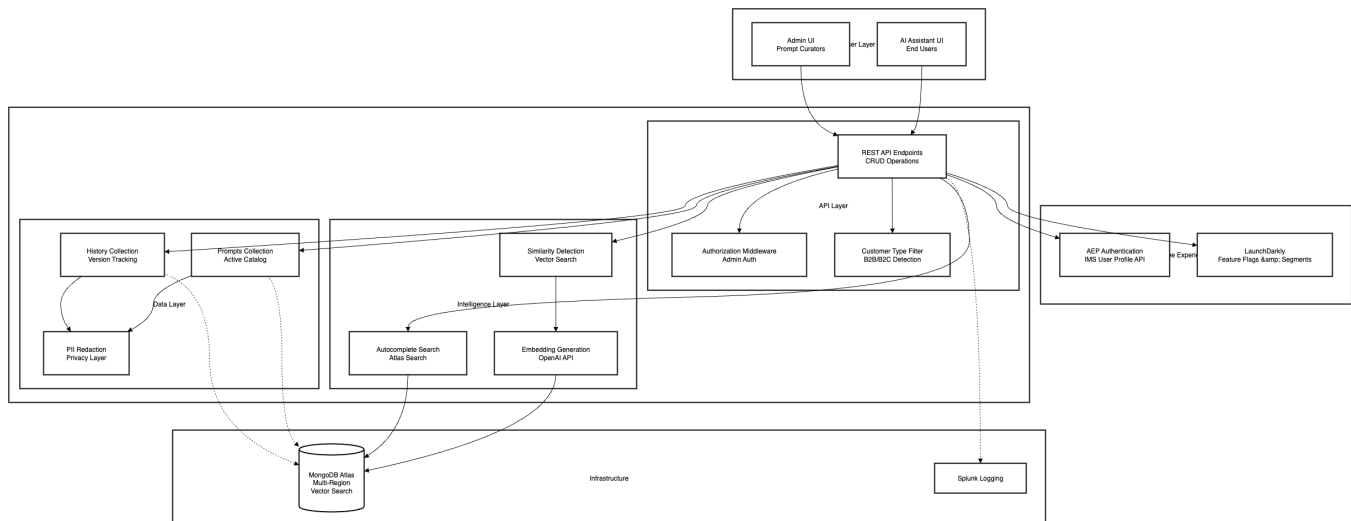
Github

Repo: <https://git.corp.adobe.com/experience-platform/agent-composer-svc> Architecture docs: <https://git.corp.adobe.com/experience-platform/agent-composer-svc/tree/main/docs> API Documentation: <https://git.corp.adobe.com/experience-platform/agent-composer-svc/blob/main/docs/promptstore/api-documentation.md>

Architecture

System Overview

The following diagram illustrates the high-level architecture and component relationships within the Prompt Library platform:



Prompt Catalog

The Prompt Catalog is the central repository that maintains structured metadata and content for all curated prompts. This catalog acts as the system of record enabling discovery, curation, and intelligent suggestion across AI Assistant experiences.

MVP Note: For the initial release, only Adobe internal administrators can manage the prompt catalog. End users have read-only access through the AI Assistant UI.

Components

- **Prompt:** Core prompt entity with text, metadata, and targeting configuration
- **Prompt History:** Version-controlled audit trail of all prompt modifications
- **Vector Embeddings:** Semantic representations for similarity detection and search
- **Admin Authorization:** LaunchDarkly-based access control for write operations
- **Customer Type Detection:** Automatic B2B/B2C filtering based on IMS Org ID

Sample Document Objects

Prompt Sample

```
{
  "_id": "6920967ddcddec594aaeb42",
  "prompt": "What can you help me with in terms of audience management?",
  "goal": ["Learning"],
  "domain": ["General"],
  "role": ["B2C Marketer"],
  "application": ["Experience Platform", "Adobe Journey Optimizer"],
  "isFeatured": true,
  "version": 2,
  "vector_embedding": [0.023, -0.145, 0.892, ...],
  "createdBy": {
    "userId": "user@adobe.com",
    "imsOrg": "org123",
    "sandboxName": "prod",
    "name": "John Doe",
    "email": "john.doe@adobe.com"
  },
  "lastModifiedBy": {
    "userId": "admin@adobe.com",
    "imsOrg": "org123",
    "sandboxName": "prod",
    "name": "Admin User",
    "email": "admin@adobe.com"
  },
  "createdAt": "2025-11-21T13:35:28.071Z",
  "updatedAt": "2025-11-22T10:00:00.000Z"
}
```

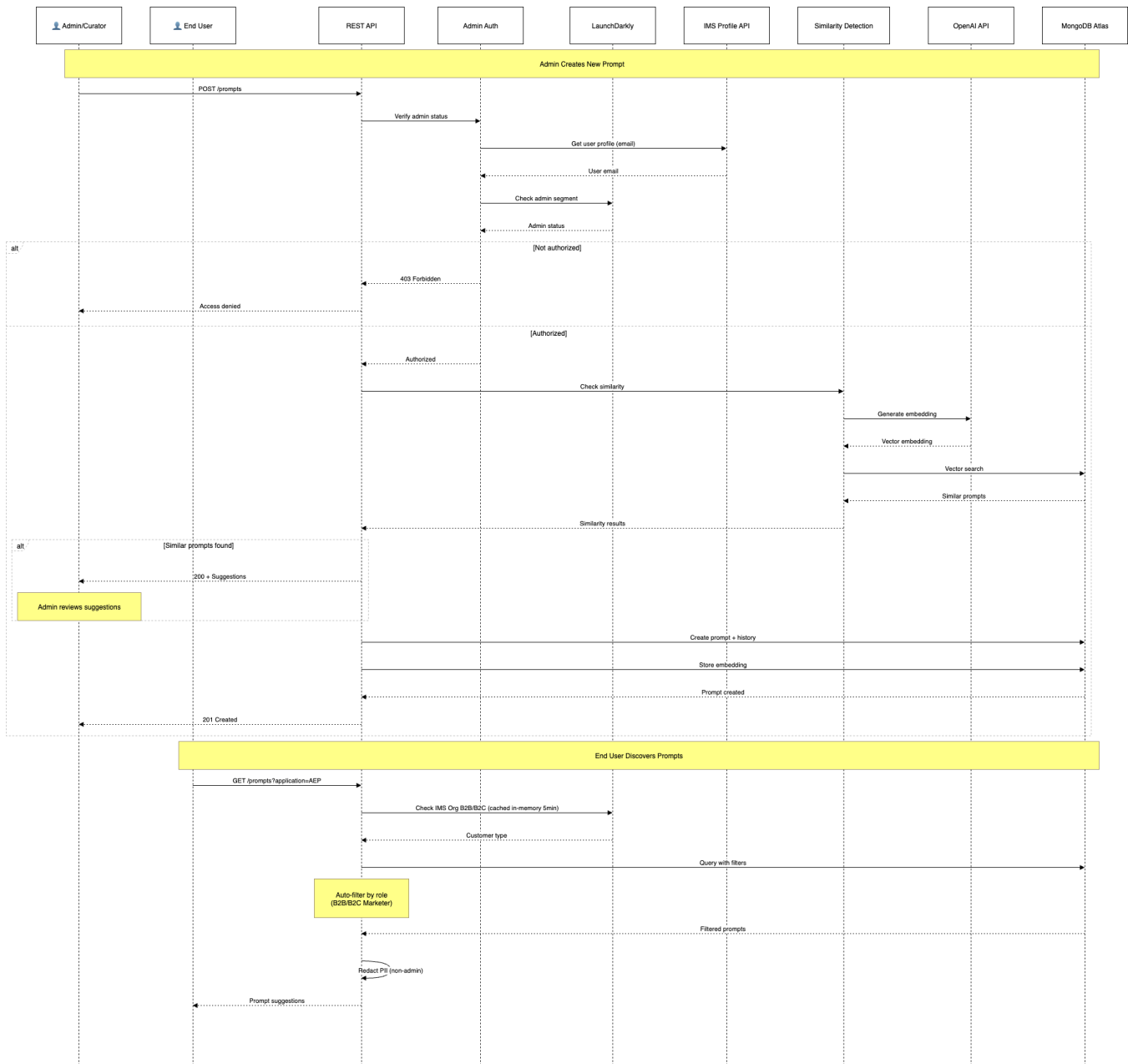
Prompt History Sample

```
{
  "_id": "hist456",
  "promptId": "6920967ddcddec594aaeb42",
  "version": 1,
  "modifiedBy": {
    "userId": "user2@adobe.com",
    "imsOrg": "org123",
    "sandboxName": "prod",
    "name": "Editor User",
    "email": "editor@adobe.com"
  },
  "modifiedAt": "2025-11-22T10:00:00.000Z",
  "changedFields": ["prompt", "goal"],
  "changesCount": 2,
  "data": {
    "prompt": "Updated prompt text",
    "goal": ["Analyze"],
    "domain": ["General"],
    "role": ["B2C Marketer"],
    "application": ["Experience Platform"],
    "isFeatured": false
  }
}
```

Access and Mutation APIs

API Sequence Flow

The following diagram shows the typical sequence of API calls and system interactions from prompt creation through end-user discovery:



The Prompt Library supports a comprehensive set of REST APIs for:

- Querying prompts with advanced filtering and search
- Creating, updating, and deleting prompts (admin-only)
- Checking similarity before creation
- Retrieving prompt history and versions
- Automatic B2B/B2C customer type filtering

API Base URLs:

- Stage: <https://agent-scheduler-api-stage.adobe.io/aia-prompt-store/v1/prompts>
- Production: <https://agent-scheduler-api.adobe.io/aia-prompt-store/v1/prompts>

Authentication: All APIs require a valid **userToken** issued by Adobe AEP in the Authorization header:

Authorization: Bearer {token}

Authorization:

- **Read Operations (GET):** Available to all authenticated users with automatic PII redaction for non-admins
- **Write Operations (POST, PUT, DELETE):** Restricted to users in the prompt-library-admins LaunchDarkly segment

Complete API Documentation: For detailed endpoint specifications, request/response formats, query parameters, error codes, and usage examples, see:

- [API Reference Guide](#) - Complete REST API documentation
- [Implementation Details](#) - Technical implementation details

Key API Endpoints:

- GET /prompts - List/search prompts with filtering
- GET /prompts/:id - Get a single prompt by ID
- POST /prompts - Create a new prompt (admin-only)
- POST /prompts/check-similarity - Check for similar prompts
- PUT /prompts/:id - Update an existing prompt (admin-only)
- DELETE /prompts/:id - Delete a prompt (admin-only)
- GET /prompts/:id/history - Get prompt change history
- GET /prompts/:id/history/:version - Get specific version

Intelligent Search & Discovery

The Prompt Library implements multiple search and discovery mechanisms to help users find relevant prompts quickly and efficiently.

Autocomplete Search

MongoDB Atlas Search Integration: Leverages Atlas Search with edge n-gram tokenization for real-time autocomplete as users type.

Search Capabilities:

- **Prefix Matching:** "aud" matches "audience", "audiences"
- **Case Insensitive:** "jour" matches "Journeys", "journal"
- **Minimum 2 Characters:** Optimized for performance and relevance
- **Relevance Scoring:** Results sorted by search score (higher = better match)

Technical Implementation:

```
// MongoDB Atlas Search Index Configuration
{
  "mappings": {
    "fields": {
      "prompt": {
        "type": "autocomplete",
        "tokenization": "edgeGram",
        "minGrams": 2,
        "maxGrams": 15,
        "foldDiacritics": true
      }
    }
  }
}
```

Performance:

- Response time: < 300ms (p95)
- No additional MongoDB indexes required
- Optimized index size through selective field indexing

Vector Similarity Search

Semantic Understanding: Uses OpenAI embeddings and MongoDB vector search for similarity detection and duplicate prevention.

Use Cases:

1. **Pre-Creation Similarity Check:** Administrators check for existing similar prompts before creating new ones
2. **Duplicate Prevention:** Automatic detection prevents catalog pollution
3. **Quality Control:** Helps maintain a clean, well-organized prompt catalog

Similarity Scoring:

- **0.95-1.0:** Nearly identical (recommend using existing)
- **0.80-0.95:** Very similar (recommend update or merge)
- **0.70-0.80:** Somewhat similar (review for overlap)
- **<0.70:** Different (safe to create)

Technical Flow:

1. User enters prompt text
2. System generates embedding via OpenAI API
3. MongoDB vector search finds similar prompts (cosine similarity)
4. Results returned with similarity scores
5. Administrator reviews and decides action

Advanced Filtering

Multi-Dimensional Filtering: Combine multiple filter types for precise prompt discovery.

Filter Types:

- **Application:** Filter by AEP product (Experience Platform, Adobe Journey Optimizer, Customer Journey Analytics)
- **Goal:** Filter by user intent (Learning, Analyze, Optimize, Troubleshooting, Create)
- **Domain:** Filter by functional area (General, Audiences, Journeys, Data Insights, Experimentation)
- **Role:** Filter by user persona (B2C Marketer, B2B Marketer, Data Engineer)
- **Featured Status:** Show only featured prompts
- **User Filter:** Find prompts modified by specific users

Filter Logic:

- **OR Logic:** Multiple values within same parameter (e.g., goal=Learning, Analyze)
- **AND Logic:** Multiple different parameters (e.g., goal=Learning&domain=Audiences)
- **Automatic Customer Type Filter:** B2B/B2C role filtering applied automatically based on IMS Org ID

Authorization & Access Control

The Prompt Library implements a sophisticated multi-layered authorization system that balances open access for discovery with strict controls for content management.

Admin Authorization

LaunchDarkly Segment-Based Access Control: Admin privileges determined by user email membership in the prompt-library-admins LaunchDarkly segment.

Authorization Flow:

1. **User Authentication:** Extract user context from Bearer token and headers
2. **IMS Profile API Call:** Retrieve user's email address (cached 1 hour)
3. **LaunchDarkly Check:** Query prompt-library-admins segment (no caching)
4. **Authorization Decision:** Grant or deny write access

Admin-Only Operations:

- POST /prompts - Create new prompts
- PUT /prompts/:id - Update existing prompts
- DELETE /prompts/:id - Delete prompts

Error Responses:

- 403 Forbidden - User not authorized to modify prompts
- 403 Forbidden - Authorization check encountered an error (fail-secure)

Fail-Secure Design: If IMS or LaunchDarkly is unavailable, access is denied rather than granted.

PII Redaction

Privacy-First Read Access: All users can query prompts, but non-admin users receive responses with PII (Personally Identifiable Information) redacted.

PII Fields Affected:

- createdBy - User who created the prompt (name, email)
- lastModifiedBy - User who last modified the prompt (name, email)
- modifiedBy - User in history records (name, email)

Redaction Behavior:

- **Admin Users:** See all fields including full user details
- **Non-Admin Users:** PII fields completely omitted from responses

Endpoints with PII Redaction:

- GET /prompts - List/search prompts
- GET /prompts/:id - Single prompt retrieval

- GET /prompts/:id/history - Prompt change history
- GET /prompts/:id/history/:version - Specific version
- POST /prompts/check-similarity - Similar prompts

Automatic Customer Type Filtering

B2B/B2C Detection: Automatically determines customer type and filters prompts by appropriate role.

Detection Flow:

1. Extract IMS Org ID from x-gw-ims-org-id header
2. Detect environment (stage/production) from origin or referer header
3. Query LaunchDarkly aep-app-b2b-customers segment (cached 5 minutes)
4. Apply role filter: B2B Marketer or B2C Marketer

Filtering Logic:

- **B2B Customers:** Automatically filter role: "B2B Marketer"
- **B2C Customers:** Automatically filter role: "B2C Marketer"
- **Combined with User Filters:** Customer type filter + user-provided role filters (AND logic)

Example:

Request: GET /prompts?role=Data Engineer
IMS Org: B2B Customer

Result: Prompts with BOTH:
- "B2B Marketer" role (required)
- "Data Engineer" role (from query)

Caching Strategy:

- LaunchDarkly responses cached in-memory for 5 minutes (using Map data structure)
- IMS profile data cached in-memory for 1 hour (using node-cache library)
- Admin segment checks not cached (immediate policy enforcement)

Version History & Audit Trail

Complete Change Tracking: Every prompt modification creates a new history record, providing a complete audit trail and rollback capability.

History Record Contents:

- **Version Number:** Incrementing version counter (starts at 0)
- **Modified By:** User who made the change (with profile information)
- **Modified At:** Timestamp of modification
- **Changed Fields:** Array of field names that were modified
- **Changes Count:** Number of fields modified
- **Data Snapshot:** Complete prompt state at that version

Use Cases:

1. **Audit Trail:** Track who changed what and when
2. **Rollback:** Restore previous prompt versions
3. **Compliance:** Demonstrate content governance
4. **Quality Control:** Review changes before approval

History API Endpoints:

- GET /prompts/:id/history - Complete change history for a prompt
- GET /prompts/:id/history/:version - Retrieve specific version

Operations and Monitoring

Logging and Observability

The Prompt Library implements comprehensive logging for operational visibility and troubleshooting. All application logs are centralized through **Splunk** for aggregation, search, and alerting capabilities.

Key Logging Areas:

- **API Request/Response:** All endpoint interactions with request metadata and response codes
- **Authorization Events:** Admin auth checks, IMS profile lookups, LaunchDarkly segment queries
- **Similarity Detection:** Embedding generation, vector search performance
- **Customer Type Detection:** B2B/B2C detection and filtering decisions

- **Error Tracking:** Failed API calls, authorization errors, database errors

Structured Logging: All logs use structured JSON format with consistent field naming for efficient querying and dashboard creation in Splunk.

Service Level Targets (SLT)

The Prompt Library defines performance and reliability targets for core system operations.

API Response Times

User-facing API endpoints maintain the following response time targets (95th percentile):

Endpoint	Target (p95)	Description
GET /prompts	< 300ms	List/search prompts with filtering
GET /prompts/:id	< 100ms	Retrieve single prompt
POST /prompts	< 1000ms	Create prompt (includes embedding generation)
PUT /prompts/:id	< 1000ms	Update prompt (includes embedding if text changed)
POST /prompts/check-similarity	< 700ms	Vector similarity search
GET /prompts/:id/history	< 200ms	Retrieve prompt history

External Service Dependencies

Prompt Library relies on several external services with the following performance expectations:

Service	Operation	Target (p95)	Fallback Behavior
OpenAI API	Embedding generation	< 500ms	Fail request (required for similarity)
LaunchDarkly	Admin segment check	< 200ms	Deny access (fail-secure)
LaunchDarkly	B2B/B2C segment check	< 200ms	Default to B2C
IMS Profile API	User profile lookup	< 300ms	Deny admin access (fail-secure)
MongoDB Atlas	Vector search	< 400ms	Return error to client
MongoDB Atlas	Text search	< 200ms	Return error to client

Caching Performance

The system implements strategic in-memory caching to optimize performance and reduce external API calls:

Cache Type	Implementation	TTL	Cache Hits Target	Benefit
IMS User Profiles	node-cache	1 hour	> 90%	Reduces IMS API calls
LaunchDarkly B2B/B2C	Map with timestamps	5 minutes	> 85%	Reduces LD API calls
LaunchDarkly Admin Segment	No cache	N/A	N/A	Immediate policy enforcement

All metrics are tracked in Splunk dashboards with automated alerting for SLT violations.

MongoDB Atlas Deployment & Utilization

The Prompt Library uses MongoDB Atlas as its primary database for storing prompts, history records, and vector embeddings. Atlas provides enterprise-grade database-as-a-service with built-in high availability, automated backups, and multi-region replication.

Deployment Configuration

Cluster Sizing:

- **Tier:** M10 (scalable vertically between M10 and M20)
- **Deployment Type:** REPLICASET (3-node replica set for high availability)
- **Storage:** 100 GB disk per cluster
- **RAM:** 2 GB per M10 instance
- **Auto-scaling:** Vertical scaling enabled between M10-M20 tiers based on workload

Regional Distribution:

Production (6 regions):

- **prod-va7:** East US 2 (Virginia) - Primary region
- **prod-nld2:** West Europe (Netherlands) - EU region
- **prod-aus5:** Australia Southeast (Sydney) - APAC region
- **prod-can2:** Canada Central (Toronto) - North America region
- **prod-gbr9:** UK South (London) - UK/EU region
- **prod-ind2:** Central India (Mumbai) - India region

Stage (1 region):

- **stage-va7:** East US 2 (Virginia) - Staging environment

Database & Collections:

- **Database:** promptstore
- **Collections:**
 - prompts - Active prompt catalog with vector embeddings
 - prompthistory - Version history and audit trail

Storage Utilization

Based on current catalog size and projected growth:

Data Type	Size per Document	Storage Impact
Prompt	~2-5 KB (with metadata)	~500 KB for 100-250 prompts
Vector Embedding	~6 KB per prompt	~600 KB for 100 prompts
History Record	~1.5-3 KB	~150 KB per 100 versions
Total Active Storage		~2-5 MB (well under 100 GB capacity)

Atlas Search Indexes:

- **textSearchIndex:** Autocomplete search with edge n-grams (2-15 characters)
- **Vector Search Index:** Cosine similarity for semantic duplicate detection

Headroom Analysis:

- **Disk capacity:** 100 GB provisioned vs ~5 MB used = ~20,000× headroom
- **Growth runway:** Current storage provides capacity for ~100,000 prompts with full history
- **Scaling trigger:** Monitor disk usage; scale to M20 if sustained usage exceeds 70 GB

High Availability & Reliability:

- **Replica Set:** 3-node configuration ensures automatic failover
- **Continuous Backups:** Point-in-time recovery with MongoDB Atlas snapshots
- **Cross-Region Replication:** Production data replicated across 6 regions for disaster recovery
- **Automated Maintenance:** Zero-downtime upgrades and patches managed by Atlas

Performance Characteristics:

- **Read/Write Performance:** M10 tier supports up to 3,000 IOPS
- **Connection Pooling:** Application uses connection pooling to optimize database connections
- **Vector Search:** Optimized for cosine similarity queries (<400ms p95)
- **Text Search:** Atlas Search provides <200ms p95 for autocomplete queries

Security & Compliance:

- **Encryption:** Data encrypted at rest and at rest (TLS 1.2+)
- **Network Security:** VPC peering with Azure networks for private connectivity
- **Access Control:** Role-based access control (RBAC) with principle of least privilege
- **Audit Logging:** Database access and operations logged for compliance

Operational Management:

- **Infrastructure as Code:** Atlas deployment managed through the exc/amplify-atlas configuration repository
- **Monitoring:** Real-time performance metrics via MongoDB Atlas dashboard and Adobe monitoring
- **Alerting:** Configured thresholds for disk usage, connection spikes, and performance degradation
- **Cost Tracking:** Monthly cost monitoring via Adobe's MongoDB cost dashboard

For cost breakdown and multi-region deployment strategy, see [Multi-Region Cost Model](#) in the COGS section

Data Retention & Governance

The Prompt Library implements a straightforward data lifecycle management strategy focused on long-term catalog retention while maintaining complete audit history.

Retention Policy Framework

The system implements a single-tier retention approach:

Data Category	Retention Period	Storage Location	Purging Policy
Prompts	Indefinite	MongoDB prompts collection	Manual deletion only (admin-initiated)
Prompt History	Indefinite	MongoDB prompthistory collection	Cascade delete when prompt deleted
Vector Embeddings	Indefinite	MongoDB prompts collection (embedded)	Deleted with parent prompt

Data Categories

Prompt Data (Retained indefinitely)

The prompts collection contains the active prompt catalog:

- **Contents:**
 - Prompt text and metadata (goal, domain, role, application)
 - Vector embeddings for similarity detection
 - Version tracking and user attribution
 - Featured status and display preferences
- **Retention:** Indefinite (manual deletion only)
- **Purging:** Only when administrator explicitly deletes prompt
- **Rationale:** Prompts are curated content requiring long-term retention for consistency and reference

Prompt History (Retained indefinitely)

The prompthistory collection contains version-controlled audit trails:

- **Contents:**
 - Historical snapshots of prompt at each version
 - Change tracking metadata (who, when, what changed)
 - Complete data state at each version
 - Modification attribution with user profiles
- **Retention:** Indefinite (tied to parent prompt)
- **Purging:** Cascade deleted when parent prompt is deleted
- **Rationale:** Audit trail required for compliance, quality control, and rollback capabilities

Data Lifecycle

Active Data Phase (Current)

- **Prompts:** Always accessible via API endpoints
- **History:** Complete version history accessible via history endpoints
- **Embeddings:** Used for real-time similarity detection
- **Storage:** Primary MongoDB collections with indexes
- **Performance:** Optimized for fast queries and real-time access

Deletion Phase (Administrator-Initiated)

When an administrator deletes a prompt:

1. Prompt document removed from prompts collection
2. All associated history records removed from prompthistory collection (cascade)
3. Vector embedding deleted with prompt
4. No soft delete or recycle bin (permanent deletion)

Deletion API:

- DELETE /prompts/:id - Hard delete (permanent)
- Returns count of history records deleted
- Cannot be undone (no rollback)

Operational Considerations

Monitoring & Alerting:

- **Storage Metrics:** Monitor prompts and prompthistory collection sizes
- **Growth Tracking:** Track document count trends over time
- **Alert Thresholds:**
 - Alert if prompts collection exceeds 1,000 documents (indicates potential spam or quality issues)

- Alert if prompt history collection exceeds 10,000 records (indicates excessive churn)
- **Performance Monitoring:** Track query performance as catalog grows

Backup & Recovery:

- **MongoDB Atlas Backups:** Continuous cloud backups provided by MongoDB Atlas
 - Point-in-time recovery available
 - Automated backup snapshots
- **Prompt Catalog Protection:** Critical curated content backed up regularly
- **History Protection:** Audit trail backed up for compliance
- **Recovery Scenarios:**
 - Database corruption: Restore from Atlas backup
 - Accidental deletion: Restore from Atlas snapshot (point-in-time recovery)
 - Disaster recovery: MongoDB Atlas handles high availability and backup redundancy

Data Privacy and Regulatory Compliance

The Prompt Library is designed with privacy-by-design principles to minimize data protection risks while ensuring compliance with global data protection regulations.

Data Processing Overview

Personal Data Collection: The Prompt Library collects limited personal data exclusively from Adobe internal administrators for audit and attribution purposes.

Data Stored:

- **User Attribution:** Creator and modifier information (name, email, user ID, IMS Org ID)
- **Purpose:** Audit trails, quality control, rollback attribution
- **Visibility:** Redacted for non-admin users, visible only to administrators
- **Source:** Adobe IMS Profile API (internal Adobe employees only)

Data Minimization: All data stored in MongoDB consists of:

- Curated prompt text and metadata (goal, domain, role, application)
- User attribution for audit purposes (admin names/emails)
- System-generated version tracking and timestamps
- Technical identifiers (prompt IDs, version numbers)

Regulatory Compliance Framework

General Data Protection Regulation (GDPR)

- **Personal Data Processing:** Limited to Adobe employee data for internal administrative purposes
- **Legal Basis:** Legitimate business interests for internal Adobe operations and quality control
- **Individual Privacy Rights:**
 - Right to access: Administrators can view their attribution via API
 - Right to erasure: Not applicable (prompt library is work product, not personal data subject to erasure)
 - Right to rectification: Administrators can update prompts (triggers new version)
- **Data Privacy:** PII automatically redacted for non-admin users (privacy-first design)

Health Insurance Portability and Accountability Act (HIPAA)

- **Health Information Processing:** System does not process or store any health-related information
- **Compliance Requirements:** Not applicable to current system scope

Data Governance and Security

Data Retention:

- **Prompts & History:** Retained indefinitely for consistency and audit purposes
- **User Attribution:** Stored permanently as part of audit trail
- **System Logs:** Managed through Splunk with configurable retention periods

Cross-Border Data Transfer:

- **Multi-Region Deployment:** Data may be processed across 6 Azure production regions (Virginia/US, Netherlands/EU, Australia, Canada, UK, India)
- **Regional Isolation:** Each region operates independently; Atlas handles cross-region replication
- **Adequacy Decisions:** Transfers comply with GDPR adequacy decisions for international data transfers
- **Standard Contractual Clauses:** Adobe's enterprise agreements include appropriate data transfer mechanisms

Privacy Impact Assessment

Risk Assessment: The Prompt Library presents minimal privacy risks due to:

- Limited personal data collection (Adobe employees only, for audit purposes)
- PII redaction for non-administrative users
- Curated content focused on business workflows, not personal information
- Strong access controls and authentication requirements

Mitigation Measures:

- **Access Control:** All data access requires valid Adobe AEP authentication
- **PII Redaction:** Automatic redaction of user information for non-admin users
- **Encryption:** Data encrypted in transit (TLS) and at rest (MongoDB encryption)
- **Audit Logging:** Comprehensive logging of all data access and modification activities
- **Admin Authorization:** LaunchDarkly segment-based access control with fail-secure design

Cost of Goods Sold (COGS)

Infrastructure Cost Components

1. MongoDB Atlas (Prompt Catalog & History)

- **Configuration:** M10 dedicated cluster (2 GB RAM, 100 GB storage), REPLICASET with 3-node replica set
- **Multi-Region Deployment:** 6-region production deployment with cross-region replication
- **Storage Model:** Based on sample document sizes:

Prompt: ~2-5 KB per document (with metadata)

Vector Embedding: ~6 KB per prompt

History Record: ~1.5-3 KB per version

- **Cost:**
 - Production: \$472/month (6-region deployment)
 - Stage: \$97/month (single-region deployment)
- **Shared with Agent Scheduler:** MongoDB costs shared across Agent Scheduler and Prompt Library platforms

2. OpenAI API (Embedding Generation)

- **Model:** text-embedding-3-small
- **Cost:** \$0.00002 per 1K tokens (~\$0.00001 per embedding)
- **Usage:** Generated on prompt creation and updates (if prompt text changes)
- **Estimated Monthly Cost:** ~\$1-5/month (based on 100-500 prompt creations/updates per month)

3. LaunchDarkly API (Admin Auth & Customer Type Detection)

- **Configuration:** Included in Adobe's enterprise LaunchDarkly subscription
- **Cost:** No direct cost (leverages existing Adobe infrastructure)
- **Usage:** Admin segment checks, B2B/B2C customer detection

4. IMS Profile API (User Profile Lookup)

- **Configuration:** Adobe internal API
- **Cost:** No direct cost (Adobe internal service)
- **Usage:** User email retrieval for admin authorization
- **Caching:** In-memory cache (node-cache) with 1-hour TTL

5. Compute Resources (Application Hosting)

- **Platform:** Adobe's internal Kubernetes infrastructure
- **Cost:** No direct cost (leverages existing Adobe infrastructure)

Cost Estimate Summary

Monthly Cost Estimate (Production - All Regions):

- **MongoDB Atlas:** \$472/month (shared with Agent Scheduler, 6-region deployment)
- **OpenAI API:** ~\$1-5/month (embedding generation)
- **LaunchDarkly:** Included in Adobe enterprise subscription
- **IMS Profile API:** No direct cost (Adobe internal)
- **Compute:** Included in Adobe infrastructure
- **Total Incremental Cost:** ~\$473-477/month (primarily MongoDB shared infrastructure)

Monthly Cost Estimate (Stage Environment):

- **MongoDB Atlas:** \$97/month (shared with Agent Scheduler, single-region deployment)
- **OpenAI API:** ~\$1/month (testing and development)

- **Total Stage Cost:** ~\$98/month

Cost per Prompt Operation:

- **Prompt Creation:** ~\$0.00001 (OpenAI embedding generation)
- **Prompt Update (text changed):** ~\$0.00001 (new embedding)
- **Prompt Update (metadata only):** \$0 (no embedding regeneration)
- **Prompt Retrieval:** \$0 (no external API calls, only MongoDB queries)
- **Similarity Check:** ~\$0.00001 (embedding generation for comparison)

Cost Efficiency Notes:

- **Shared Infrastructure:** MongoDB costs shared with Agent Scheduler platform
- **Minimal External API Costs:** OpenAI API usage is negligible (~\$1-5/month)
- **In-Memory Caching Strategy:** 5-minute and 1-hour in-memory caches minimize LaunchDarkly and IMS API calls
- **No Per-User Costs:** Unlimited users with fixed infrastructure costs

Storage Assumptions

- Average prompt size: ~2-5 KB per prompt (with metadata)
- Vector embedding: ~6 KB per prompt
- History record: ~1.5-3 KB per version
- Expected catalog size: 100-500 prompts at launch
- Expected history records: 500-2,000 versions (multiple versions per prompt)
- Total storage: ~2-10 MB (well under MongoDB 100 GB capacity)

See [MongoDB Atlas Deployment & Utilization](#) for detailed configuration and storage analysis

Cost Optimization Opportunities

OpenAI API Optimization:

- Current embedding model (text-embedding-3-small) is cost-optimized
- Embedding generation only on prompt text changes (not metadata updates)
- **Potential optimization:** Batch embedding generation for bulk imports (future enhancement)

Caching Optimization:

- LaunchDarkly responses cached in-memory 5 minutes (reduces LD API calls by ~90%)
- IMS profiles cached in-memory 1 hour (reduces IMS API calls by ~95%)
- **Current state:** Already optimized; no further improvements needed

Multi-Region Deployment

The Prompt Library shares the same multi-region infrastructure as the Agent Scheduler platform. Each region operates independently with MongoDB Atlas providing automatic cross-region replication.

Regional Distribution

Production Regions (6 regions):

- **prod-va7:** East US 2 (Virginia) - Primary US region
- **prod-nld2:** West Europe (Netherlands) - Primary EU region
- **prod-aus5:** Australia Southeast (Sydney)
- **prod-can2:** Canada Central (Toronto)
- **prod-gbr9:** UK South (London)
- **prod-ind2:** Central India (Mumbai)

Non-Production Regions:

- **stage-va7:** East US 2 (Virginia) - Staging environment
- **preprod-va7:** East US 2 (Virginia) - Pre-production environment
- **dev-va7:** East US 2 (Virginia) - Development environment

Infrastructure Architecture

MongoDB Atlas: Multi-region replica set deployment (shared with Agent Scheduler)

- **Production:** 6-region replica set (va7, nld2, aus5, can2, gbr9, ind2)
- **Stage:** Single-region deployment (va7)
- Not deployed through Terraform - managed via the `exc/amplify-atlas` configuration repository
- Atlas handles automatic replication, failover, and cross-region data synchronization
- Connection details managed via vault
- M10 tier with 100 GB storage and 3-node replica set per region
- **Database:** promptstore (dedicated database for Prompt Library)

- **Collections:** prompts, prompthistory

Deployment Strategy

- Each region serves users from that geographic area for optimal performance
- MongoDB Atlas provides automatic cross-region replication and failover
- Users connect to their nearest region for lowest latency
- Regional failures handled by MongoDB Atlas automatic failover

For multi-region cost breakdown and total infrastructure costs, see [Multi-Region Cost Model](#) in the COGS section

Post-MVP / Future Enhancements

The following capabilities represent potential future enhancements to the Prompt Library platform. These features are not part of the initial MVP release but have been identified as valuable additions that could significantly enhance prompt discovery, quality control, and user experience.

Usage Analytics & Insights

The Prompt Library could be enhanced with usage analytics to track prompt effectiveness and user engagement patterns.

Use Case: Prompt Performance Tracking

Current State: Administrators manage prompts without data on which prompts are actually used or effective.

Enhanced Capability: Track prompt clicks, usage frequency, and user engagement to inform curation decisions.

Metrics to Track:

- **Click-through Rate:** How often users click on suggested prompts
- **Completion Rate:** How often clicked prompts result in successful AI interactions
- **User Feedback:** Thumbs up/down on prompt suggestions
- **Popular Prompts:** Most-used prompts by application, role, and domain
- **Abandoned Prompts:** Prompts that are never used (candidates for removal)

Benefits:

- Data-driven prompt curation decisions
- Identify high-value prompts for featured promotion
- Remove low-engagement prompts to reduce clutter
- A/B testing of prompt variations

Prompt Localization

The Prompt Library could be enhanced with multi-language support for global user base.

Use Case: Multilingual Prompt Catalog

Current State: All prompts are in English only.

Enhanced Capability: Support prompts in multiple languages with automatic user locale detection.

Implementation Approach:

- Store prompt translations in additional language fields
- Detect user locale from browser settings or user profile
- Return prompts in user's preferred language
- Admin UI for managing translations
- Automatic translation suggestions via AI for rapid localization

Supported Languages (Potential):

- English (en-US)
- French (fr-FR)
- German (de-DE)
- Japanese (ja-JP)
- Chinese Simplified (zh-CN)

Prompt Recommendation Engine

The Prompt Library could be enhanced with an AI-powered recommendation engine that suggests prompts based on user behavior and context.

Use Case: Intelligent Prompt Suggestions

Current State: Prompts displayed based on static filters (role, application, domain).

Enhanced Capability: Machine learning model that personalizes prompt suggestions based on:

- User's recent queries and interactions
- Similar users' behavior (collaborative filtering)
- Current page/context in application
- Time of day, day of week patterns
- User's role and historical preferences

Implementation Approach:

- Capture user interaction data (prompt clicks, chat submissions)
- Train collaborative filtering model on user-prompt interactions
- Real-time inference for personalized ranking
- A/B testing to measure recommendation quality

Benefits:

- Improved prompt discovery (users find relevant prompts faster)
 - Higher engagement rates (more prompt clicks and usage)
 - Reduced time-to-insight (users get to answers faster)
 - Continuous improvement through ML model training
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References

API Documentation

- Prompt Library API Reference: <https://git.corp.adobe.com/experience-platform/agent-composer-svc/blob/main/docs/promptstore/api-documentation.md>
- API Implementation Details: <https://git.corp.adobe.com/experience-platform/agent-composer-svc/blob/main/docs/promptstore/api-implementation.md>

Related Architecture Documents

- Agent Scheduler Architecture: <https://git.corp.adobe.com/experience-platform/agent-composer-svc/blob/main/docs/architecture.md>

LaunchDarkly

- LaunchDarkly Segments: <https://app.launchdarkly.com/>
- Feature Flag Management: <https://app.launchdarkly.com/projects/unified-shell/flags/prompt-library-admin-ui/targeting?env=production&env=stage&selected-env=stage>

MongoDB Atlas Configuration & Cost Tracking

Production Cost Dashboard:

- <https://cjm.gf.corp.adobe.com/d/bebfsm9ofteyod/mongo-costs?orgId=1&var-product=AJO&var-environment=All&var-cluster=All&var-topN=10&var-aggrInterval=1M>

Stage Cost Dashboard:

- <https://cjm.gf.corp.adobe.com/d/bebfsm9ofteyod/mongo-costs?orgId=1&var-product=AJO&var-environment=All&var-cluster=All&var-topN=10&var-aggrInterval=1M>

Infrastructure as Code - Amplify Atlas Configuration:

- Atlas Organization & Project Config: <https://git.corp.adobe.com/exc/amplify-atlas/blob/main/atlas-project-config/values/projects/jacs.yaml>
- Database: promptstore (dedicated database for Prompt Library)

Deployment Details:

- **Production:** 6-region replica set (va7, nld2, aus5, can2, gbr9, ind2) - Shared with Agent Scheduler
- **Stage:** Single-region deployment (va7) - Shared with Agent Scheduler
- **Cluster Tier:** M10 (scalable between M10-M20)
- **Storage:** 100 GB per cluster
- **Deployment Type:** REPLICASET with 3-node replica set for high availability